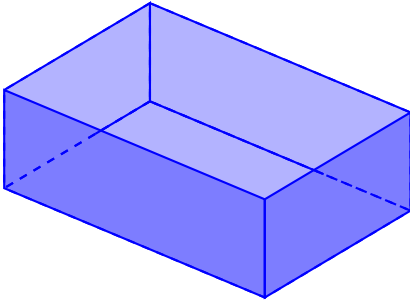


Example 1. A rectangular box with no lid is made using 48 cm^2 of cardboard. Find the box dimensions that maximize volume, assuming they exist.



Solution. First we introduce variables by labelling our sketch.

The function we wish to maximize is

The surface area is

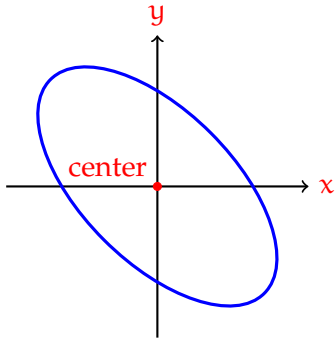
The constraints are

Let us solve the Lagrange system of equations.

Lagrange multipliers will succeed because we are told to assume there exist dimensions that maximize volume, and further those dimensions will not be degenerate, in the sense that they will not yield $\vec{\nabla}A = \vec{0}$, as the positivity of x, y, z will make all entries of $\vec{\nabla}A$ positive.

Example 2. Find the circle of largest radius centered at the origin that can be inscribed in the ellipse $x^2 + xy + y^2 = 3$.

Assume that there is a nondegenerate optimizer for your optimizing function.



Solution. The given assumption implies that Lagrange multipliers will succeed.

The constraint function is:

The radius of the largest ellipse–inscribed circle centered at the origin will match

Therefore we wish to minimize distance of (x, y) to the origin

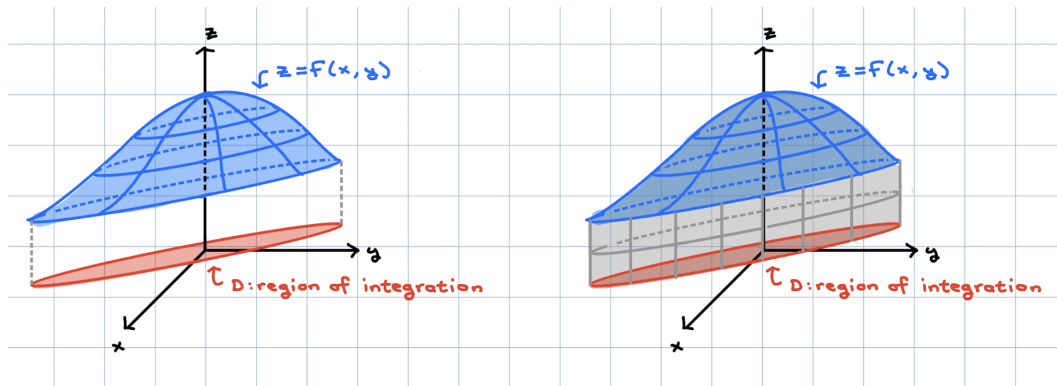
To locate the closest point on the ellipse to the origin alternatively we minimize

We set up the Lagrange multipliers system of equations.

We evaluate to find the minimum distance.

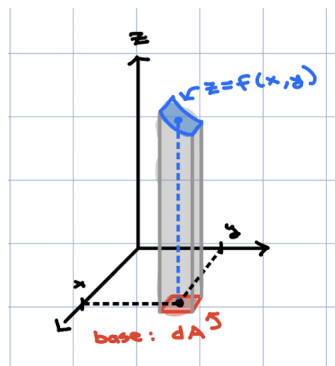
A. **Double Integrals.** Let's use integrals to calculate volumes.

Let $z = f(x, y)$ be graph and D be a **region of integration** in the xy -plane.



We will calculate the

We decompose this volume up into **infinitesimally** thin rectangular boxes.



Each infinitesimally thin rectangular box has

[base area] =

[height] =

[volume] =

The notation that produces an infinite sum of infinitesimal quantities is the:

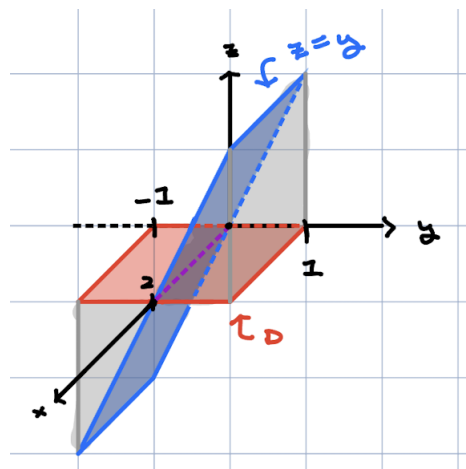
Because f is a function of two variables, the notation is called a:

We write it as follows.

[signed volume between graph of f and D] =

Example 3. Let D be the rectangle $0 \leq x \leq 2$ and $-1 \leq y \leq 1$, which we denote

$D =$



Find $\iint_D y \, dA$.

Solution. This double integral equals

The sketch shows

Hence: